Longevity orthologs across four major model organisms

Complete visual catalogue • 320 genes • 206 direct pairs

Reviewed example = a checked experiment; GenAge annotation = catalogue evidence only.

All genes in the qualifying pair set are included; rows organise direct links, not proven conserved mechanisms.

	Budding yeast	Nematode	Fruit fly	Mouse
C027 **SAC1** 2 genes / 2 species 1 direct pair	**Y1 SAC1** GenAge 1501 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 18340043	**W1 sac-1** GenAge 248 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 18006689	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				
C074 **SAM1** 3 genes / 2 species 2 direct pairs	**Y1 SAM1** GenAge 1660 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 18340043	**W1 sams-1** GenAge 584 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 16103914 **W2 sams-3** GenAge 903 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 18006689	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (n:m); W2–Y1 (n:m)				
C075 **SDH2** 2 genes / 2 species 1 direct pair	**Y1 SDH2** GenAge 2136 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 26456335	No gene in this candidate network	**F1 SdhB** GenAge 593 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 17056719	No gene in this candidate network
Direct pairs: F1–Y1 (1:1)				
C030 **SIP2** 2 genes / 2 species 1 direct pair	**Y1 SIP2** GenAge 603 **GenAge annotation** **Unclear** Lifespan: Increase + Decrease Refs: 10921902, 19030232 20421943, 21906795	**W1 aakb-1** GenAge 269 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 16673436	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.
GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.
Sources: GenAge Build 21; OMA All.May2026. See atlas legend, gene index and source tables for evidence details.